

Photosynthesis is a process used by plants to make food using sunlight.

It happens in the chloroplasts of plant cells where chlorophyll is present.

The equation for photosynthesis is:

$\text{CO}_2 + \text{H}_2\text{O} + \text{sunlight}$ gives glucose and oxygen

There are two stages in photosynthesis. The first stage uses light to produce energy molecules ATP and NADPH. Water is broken down and oxygen is released as a byproduct during this stage.

The second stage is called the Calvin cycle. In this stage carbon dioxide is used along with ATP and NADPH to make glucose. This happens in the stroma of the chloroplast.

Factors like light, CO_2 levels and temperature affect how fast photosynthesis happens.